Nearest Neighbors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Nearest Neighbors

> Prepare Nearest Neighbors: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Compute absolute **distance**s and rank **training neighbour**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/22, ai-4/12. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Write an observable label rule with boundaries and a review option

Reference example, not the only acceptable wording: Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible.

A partly hidden lane requires review because clear cannot be established from the available description.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

Concept to explain

k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.

Term	Meaning
distance	numeric measure of dissimilarity
neighbor	nearby labeled training example
k	number of neighbors voting
scaling	make feature ranges comparable

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Use a single numeric feature to make **distance** arithmetic visible, then distinguish **neighbour** selection from voting. Store only **training** rows.

Alphabetical IDs settle equal **distance**s for this classroom method; explain that it is a convention and other implementations can use different tie behavior.

Readiness answer

It stores a root question and leaf **prediction**s learned from **training**. A new example follows the appropriate branch to a stored leaf **prediction**.

Printed lesson and workbook

Number line from 0 to 14 cm and five labelled **training** markers

Optional Python 3 editor with no packages required

Blank table for **distance**s ordered IDs **vote**s and **prediction**s

End goal for the pictorial case

By the end, I can explain nearest **neighbor**s using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Nearest Neighbors using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Calculate **distance**s and compare k=1 with k=3.

MODEL IDEA k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD New leaf has normalized features (2, 3). Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 New leaf has normalized features (2, 3).

Step 2

STEP 2 Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

Step 3

STEP 3 With k=1, nearest label wins; with k=3, majority among all three wins.

Step 4

STEP 4 Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

The teaching case

Find k=1 **neighbour**s and handle ties.

Exact explanation to model

Why the result follows

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Explain the added worked example

Explain the added worked example · 1

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

Explain the added worked example · 2

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

Explain the added worked example · 3

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

If k were 1 under the same **distance** tie rule, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

Explain the added worked example · 4

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

Key terms

Term	Meaning
Distance	A numerical measure of difference between selected feature values
Neighbour	A stored training example close under the chosen distance
k	The number of neighbours selected for a prediction
Vote	A contribution of a selected neighbour’s label to the predicted class

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Model the reasoning aloud · 5

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

> **Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?**

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Nearest Neighbors

Let a nearest-**neighbour** picture show each **distance** before voting.

The facts we will use

Supplied facts

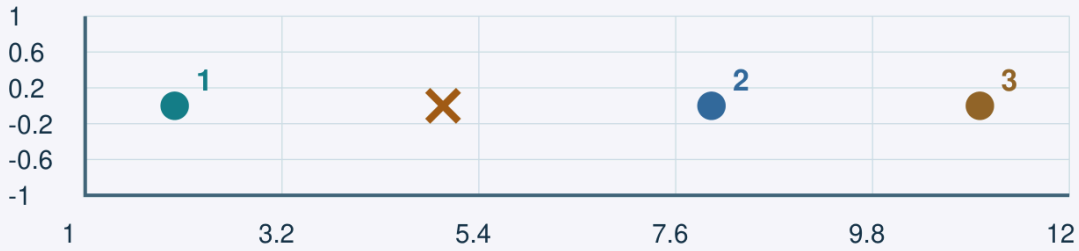
Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. **Distance** is absolute difference; ties use alphabetically earlier ID.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/12

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.



2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.



3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.
2. Apply k=1 rule	A and B tie nearest; declared ID tie-break chooses A, predicting healthy.
3. Compare k=3	All three vote: two spotted and one healthy, so k=3 predicts spotted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Length alone is a toy feature, not a disease diagnosis. **Distance** choices and feature **scaling** can change **neighbour**s.

Change one thing in the pictured case

Change one thing

Change: Move the query to length7 cm.

Predicted result

Prediction: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Why the result follows

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Nearest Neighbors, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

2. State the decision to explain

Write the question to explain: Find k=1 **neighbour**s and handle ties.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Distance**s 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Hold back the independent task

Keep for an independent attempt: For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: abs(2-3) and abs(4-3) both equal 1. For k=1 alphabetical tie order chooses A, blocked. k=3 adds B and C, whose clear **vote**s win 2 to 1. ID order is a deterministic convention, not **evidence** of the query's true label.

Exit: using labelled validation examples in the **neighbour** pool leaks evaluation answers and may let a query select itself. Restore the original **training**-only pool and score every validation query without adding its label.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve tree storage and prediction. Introduce prediction from stored examples with a declared distance.
7 to 18 minutes	Place A through E on the number line. Calculate absolute distances to query 3 and complete W1.

Stage	Teacher action
18 to 32 minutes	Trace tuple sorting and list slicing. Complete W2; distinguish tied distances from a tied label vote.
32 to 45 minutes	Read the diagram and answer the guided question. Explain why ID order is not evidence of truth.
Session 2 start to 8 minutes	Retrieve the training-only neighbour pool. Ask why validation labels must remain outside it.
8 to 20 minutes	Compute the three validation predictions and complete W3. Compare the fixed training-majority baseline.
20 to 35 minutes	Collect W4 independently before support. Require ordered IDs, both k predictions and the full validation score.
35 to 45 minutes	Use W5 and the exit question. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Completion practice

Completion task: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Reasoning to listen for

Teacher check: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID tie rule stay the same; the number of voting **neighbour**s changes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Collect explanations, not only yes/no **vote**s. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

A healthy and B spotted are equally near. The declared alphabetical rule resolves the otherwise ambiguous one-**neighbour** selection. Changed-case check: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Distances become 5, 1, 4 cm, so the nearest tie disappears.

Changed-case reasoning: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Nearest Neighbors investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. **Neighbor** Map: compute **distance**s, rank **neighbor**s, compare k values, and show how **scaling** changes one result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Nearest **neighbor** means the photo that looks most similar to a person.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A negative difference is a negative **distance**: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Odd k prevents every kind of tie: Three binary **vote**s avoid a label tie, but **neighbour**s can still tie in **distance**, including at the selection boundary.

An exact match proves the label: It matches the selected feature only. Other conditions, annotation errors or contradictory examples can remain.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Nearest Neighbors questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Independent question

Independent question · withhold the key

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

Expected independent answer

Teacher key · Expected independent answer

Distances are 4 to fern and 2 to grass, so predict grass. This result assumes the specified feature and **distance** rule.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Nearest-**neighbor prediction** depends on the order examples are listed.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Distances are 6,2,2, producing conflicting nearest labels. Use a predefined tie policy or abstain.

Raw numeric **distance**s depend on units; standardise compatible measurements and fit any learned **scaling** using **training** data only.

Explanations for the additional workbook examples

W1 Calculate feature distances

Workbook question · W1

For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

Teacher reasoning · W1

Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the **distance** calculation.

W2 Apply neighbour and tie rules

Workbook question · W2

For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID

order on equal **distance**s. Explain what changes and what remains the same.

Teacher reasoning · W2

For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID tie rule stay the same; the number of voting **neighbour**s changes.

W3 Keep evaluation examples separate

Workbook question · W3

Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

Teacher reasoning · W3

Adding validation labels uses evaluation answers to make **prediction**s and changes the intended comparison with unseen trips. The fixed k=3 method predicts clear three times and gets 1/3.

The **training**-majority clear baseline also gets 1/3.

W4 Find neighbours independently

Workbook question · W4

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

Teacher reasoning · W4

Query 6 **distance**s P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. k=1 predicts blocked from Q; k=3 uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R.

All predict blocked, matching validation rows 1 and 3: 2/3. **Training**-majority blocked also scores 2/3.

W5 Interpret a vote and a match

Workbook question · W5

Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

Teacher reasoning · W5

The **vote** fraction describes the selected **training** labels, not a verified probability of correctness. An exact match on one feature can still differ in omitted conditions or have a wrong or conflicting label.

Separate evaluation and relevant features remain necessary.

Assess independent transfer

Independent question · withhold the key

Selected case: **Training** points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute **distance**; for a nearest-**neighbor** tie choose the alphabetically earlier ID. Compare k=1 and k=3.

Selected case answer

> **Teacher key · Selected case answer**
>
> **Distance**s A6,B2,C2. The specified tie rule selects B for k=1, predicting spotted. With k=3, two healthy labels out**vote** one spotted. State the **distance** and tie policy; this tiny case establishes no wider accuracy.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Nearest Neighbors to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 13: Train a Classifier. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use physical steps on a number line before abs. Supply an empty **distance** table without the order.

Extension: explain how renaming tied IDs can change a **prediction** under this policy while the measured features remain unchanged, and why the policy must be recorded.

Require W4 **distance**s 5,1,1,5,7, order Q,R,P,S,T, both **prediction**s blocked, and validation plus baseline scores 2/3. Fresh retry: L at 0 blocked, M at 4 clear, N at 6 clear, O at 10 blocked, P at 12 clear; query 5.

Distances are 5,1,1,5,7; order M,N,L,O,P. k=1 is clear from M; k=3 is clear from M,N,L. Ask why L wins the third-place **distance** tie over O: alphabetical ID policy. Require a **training**-only pool and no probability claim.

Week 11 covers **scaling** without leakage. Carry forward **training**-only storage and explicit **distance** choices.

Teacher reference: https://scikit-learn.org/stable/modules/**neighbor**s.html (nearest-**neighbour** classification and tie considerations; reviewed 2026-09-27).

Plan the delayed return

Suggested return points in this level: ai-4/13, ai-4/15, ai-4/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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Prepare, teach and assess

> Use this preparation summary alongside the detailed Nearest Neighbors lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Compute absolute **distance**s and rank **training neighbour**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Supported practice answer

Reasoned practice answer: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID tie rule stay the same; the number of voting **neighbour**s changes.

Check every learner before discussion: Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Move the query to length7 cm. Require a **prediction** and a reason before comparing. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

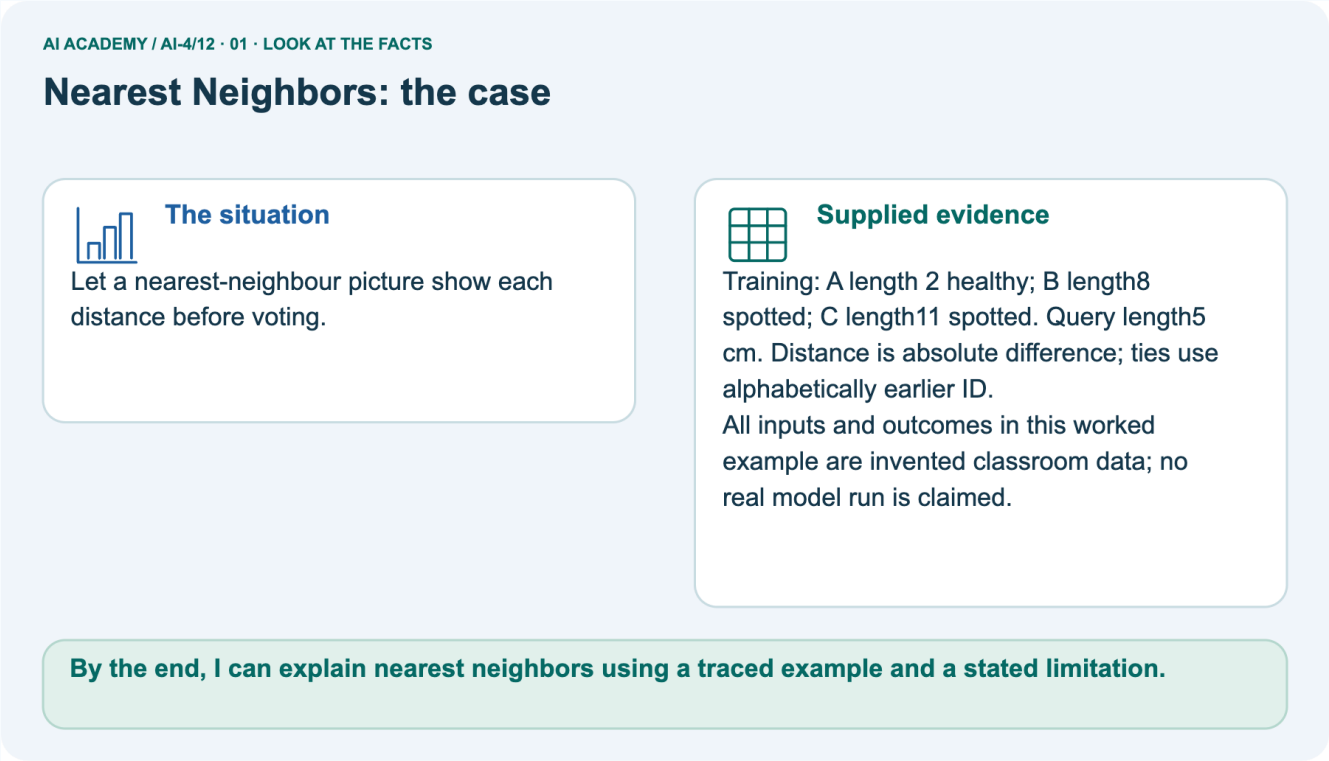
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Visual concept journey

> Show one Nearest Neighbors graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Nearest Neighbors: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Let a nearest-**neighbour** picture show each **distance** before voting. Supplied facts: **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.

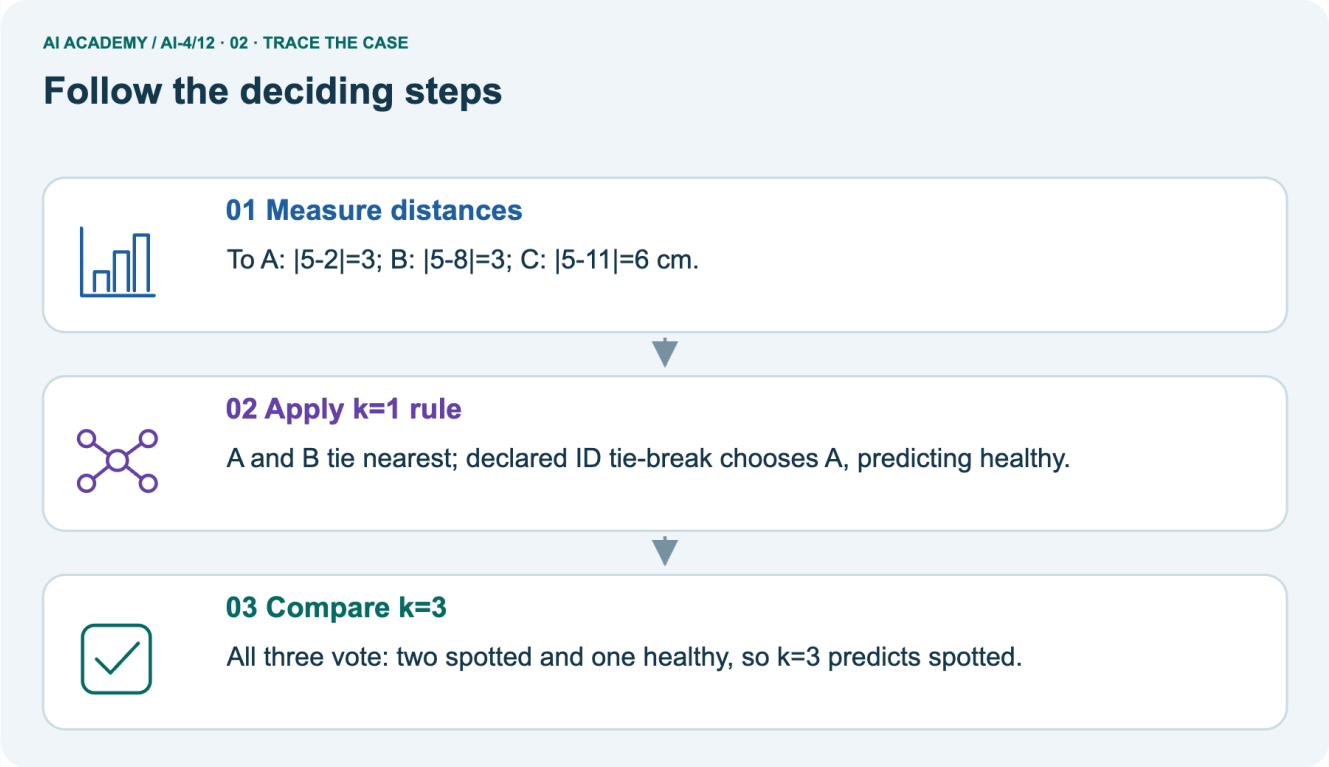
Distance is absolute difference; ties use alphabetically earlier ID. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

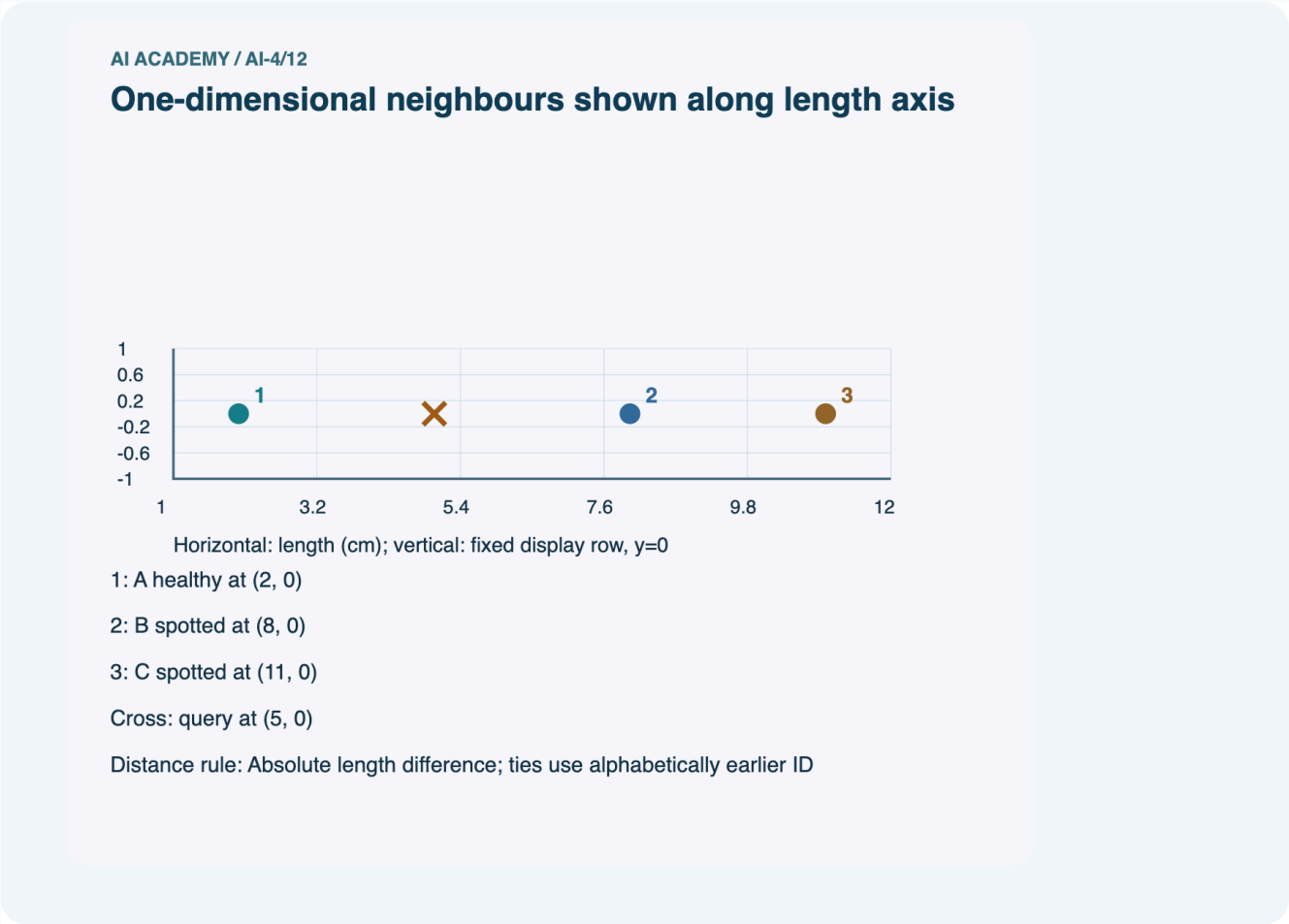
Measure **distance**s: To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm. → Apply k=1 rule: A and B tie nearest; declared ID tie-break chooses A, predicting healthy. → Compare k=3: All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One-dimensional neighbours shown along length axis



Follow the labelled picture with the matching explanation. Open the vector diagram

One-dimensional **neighbour**s shown along length axis. A healthy: (2, 0). B spotted: (8, 0). C spotted: (11, 0). Query query: (5, 0). Axes: length (cm); fixed display row, y=0.

Distance: Absolute length difference; ties use alphabetically earlier ID.

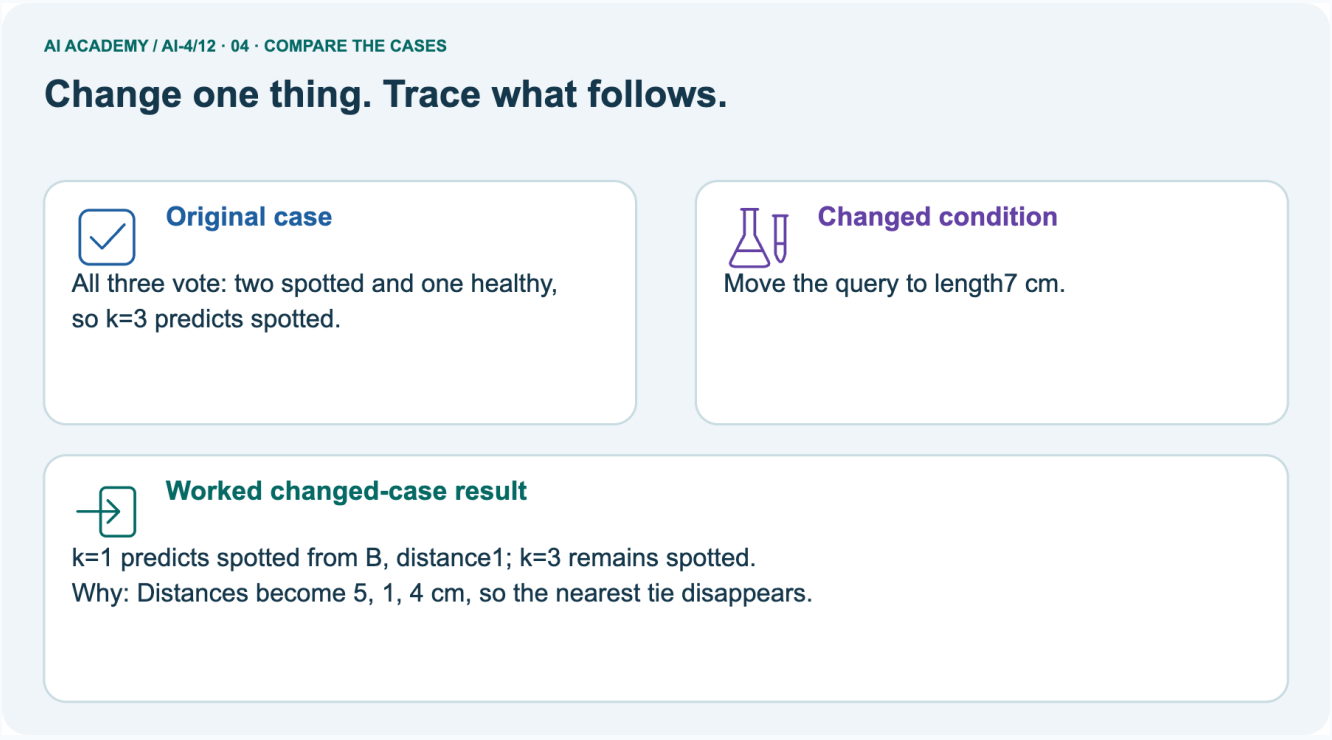
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Changed condition: Move the query to length7 cm. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

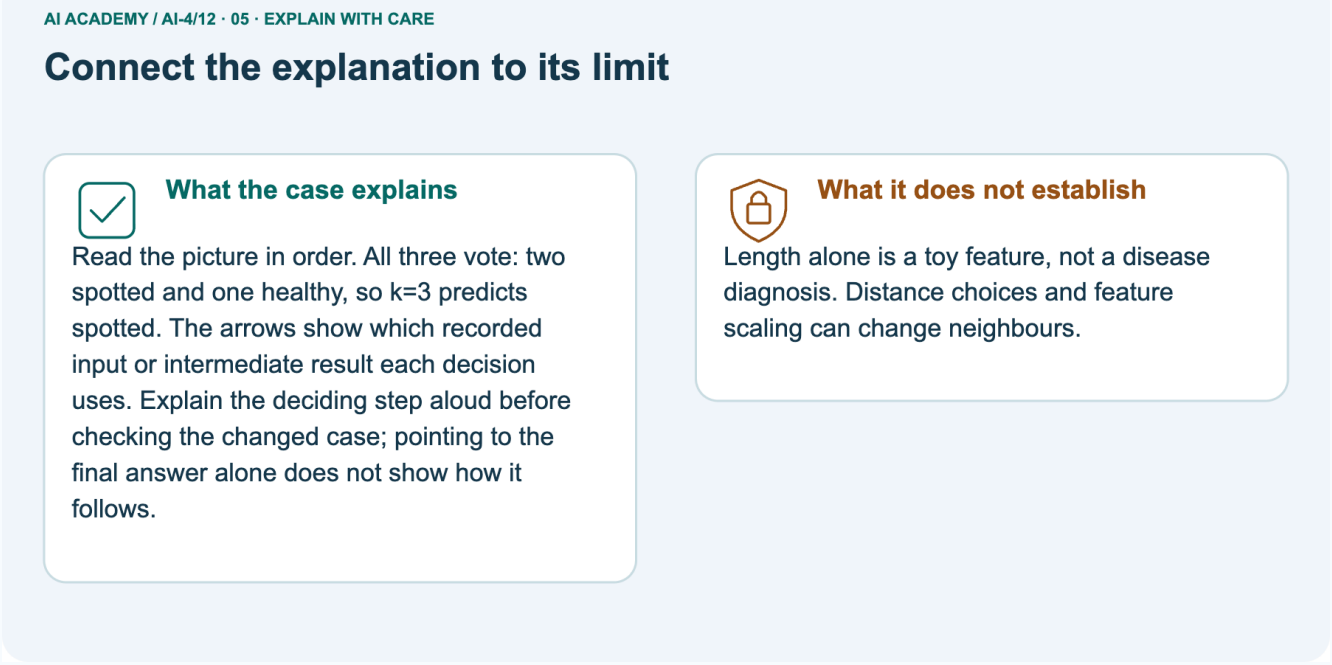
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Length alone is a toy feature, not a disease diagnosis.

Distance choices and feature **scaling** can change **neighbour**s.

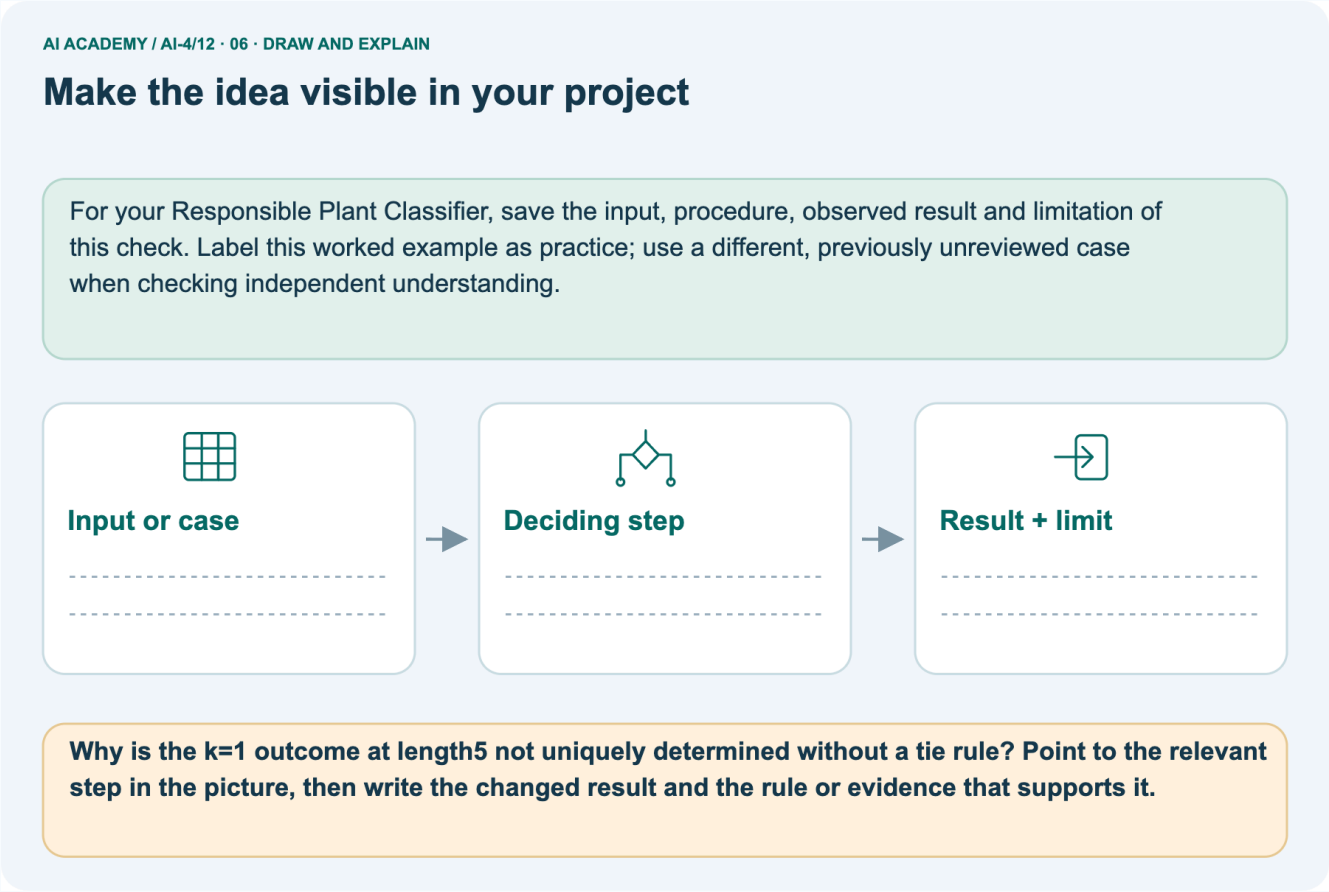
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Length alone is a toy feature, not a disease diagnosis. **Distance** choices and feature **scaling** can change **neighbour**s.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A healthy and B spotted are equally near. The declared alphabetical rule resolves the otherwise ambiguous one-**neighbour** selection.

Changed-case check: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.